

Pre-Week Homework 04

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"For any integers m and n , if m and n are both even then their sum is even."

Assume m and n are even. If $m \in \mathbb{Z}$ and $n \in \mathbb{Z}$, then since $m = 2a$ and $n = 2b$ would have the sum $m + n = 2a + 2b$, there would be an integer $k \in \mathbb{Z}$ that equals $2a + 2b$ which is divisible by 2. This makes the sum of m and n even.